ELISHA AWITI MATANGA

Python & Machine Learning Engineering | Production Data Systems

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PROFESSIONAL PROFILE

Software engineer with approximately seven years across software development, technical delivery and technology operations, with applied transaction-risk scoring, data-quality pipelines and model evaluation experience. Builds production-oriented Python services, auditable data workflows and secure APIs; pursuing an MSc in Data Science. Brings hands-on engineering, cross-functional delivery and technical leadership. Experience in full-time professional ML development should be assessed separately from broader software engineering tenure.

TECHNICAL SKILLS

Python, FastAPI, Django, Pandas/NumPy and scikit-learn concepts; SQL, PostgreSQL, SQLAlchemy, Alembic; ML evaluation (PR-AUC, ROC-AUC, calibration, threshold and cost analysis); data validation, scoring pipelines, reconciliation, model lineage; Docker, Kubernetes, Azure, CI, Pytest, OIDC/JWKS, REST APIs. Deep-learning frameworks, NLP and computer vision: developing areas, not claimed as production expertise.

RELEVANT APPLIED ML / DATA PROJECT

Transaction Intelligence Lab | Python, FastAPI, PostgreSQL, Docker, Pytest

- Built an auditable transaction-risk platform with ingestion, data-quality quarantine, deterministic baseline scoring and prediction lineage using synthetic data.
- Implemented evaluation workflows covering PR-AUC, ROC-AUC, log loss, Brier score, calibration, threshold sweeps and cost-sensitive analysis.
- Engineered reconciliation, model-artifact validation, controlled promotion/rollback, secure OIDC/JWKS access, health checks and automated tests.
- Designed for reproducibility and production readiness; not represented as an institutionally deployed trained ML model.

PROFESSIONAL EXPERIENCE

Backend Engineer — Equity Group | Oct 2024–Present | Kenya

- Own and improve 57 API integrations serving 28 USSD services across six subsidiaries at millions-of-request scale.
- Support digital-banking workflows, production troubleshooting, secure releases and cross-team operational requirements.
- Implemented reusable shared logic and country-aware opt-in/opt-out controls with backward-compatibility safeguards.

Software Engineer (Remote) — Carswafi | Jan 2024–Present | Kenya

- Lead hands-on engineering of a multi-service marketplace and operator platform using Django REST, PostgreSQL, Celery and Next.js.
- Design reliable asynchronous processing, real-time request lifecycles, auditability, authorization and operational event delivery.

Software Engineer & Technical Lead — E & M Technology Company Ltd | Jan–Nov 2024 | Kenya

- Led technical delivery for a 21-developer team building a multi-tenant SACCO platform adopted by more than 10 organizations.
- Translated operational needs into architecture, engineering work, testing and release decisions.

Operations & Technical Field Support — Equity Bank Kenya Ltd | Oct 2022–Jan 2024

Supported digital banking, customer onboarding and troubleshooting with technical and network operations teams.

Product Developer — Jenga Customer | Jan 2020–2023

Designed digital-wallet journeys, transaction workflows, data models and integration concepts; Kenyan copyright registration.

Software Developer — Eletra Software Company Ltd | Oct 2019–Jan 2020

Contributed to Python/Java development, SQL applications, testing and client-facing technical work.

EDUCATION

MSc Data Science — University of Nairobi | Sep 2025–Expected Sep 2027

BSc Computer Science — Maasai Mara University | 2017–2021

LANGUAGES

English | Kiswahili